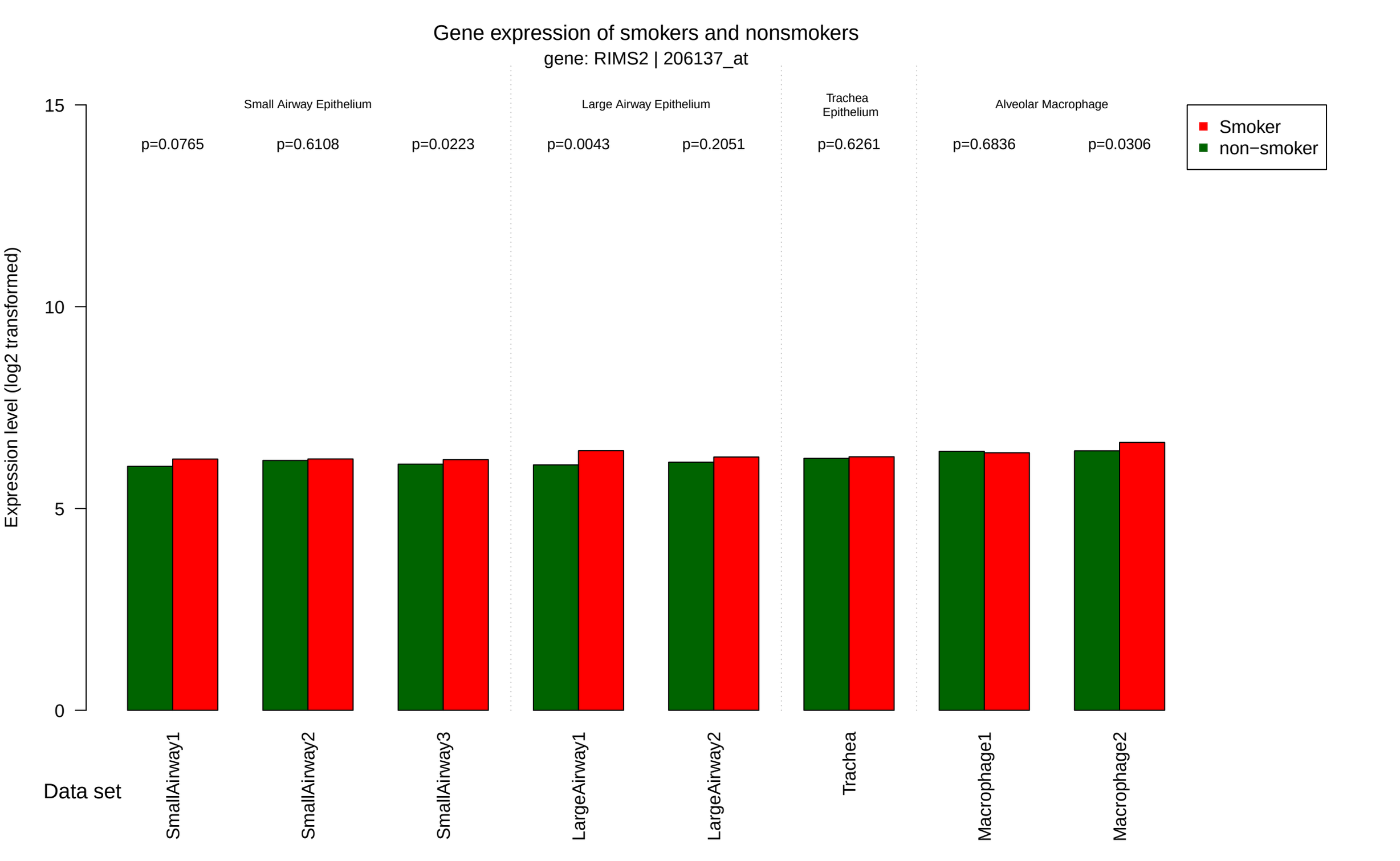
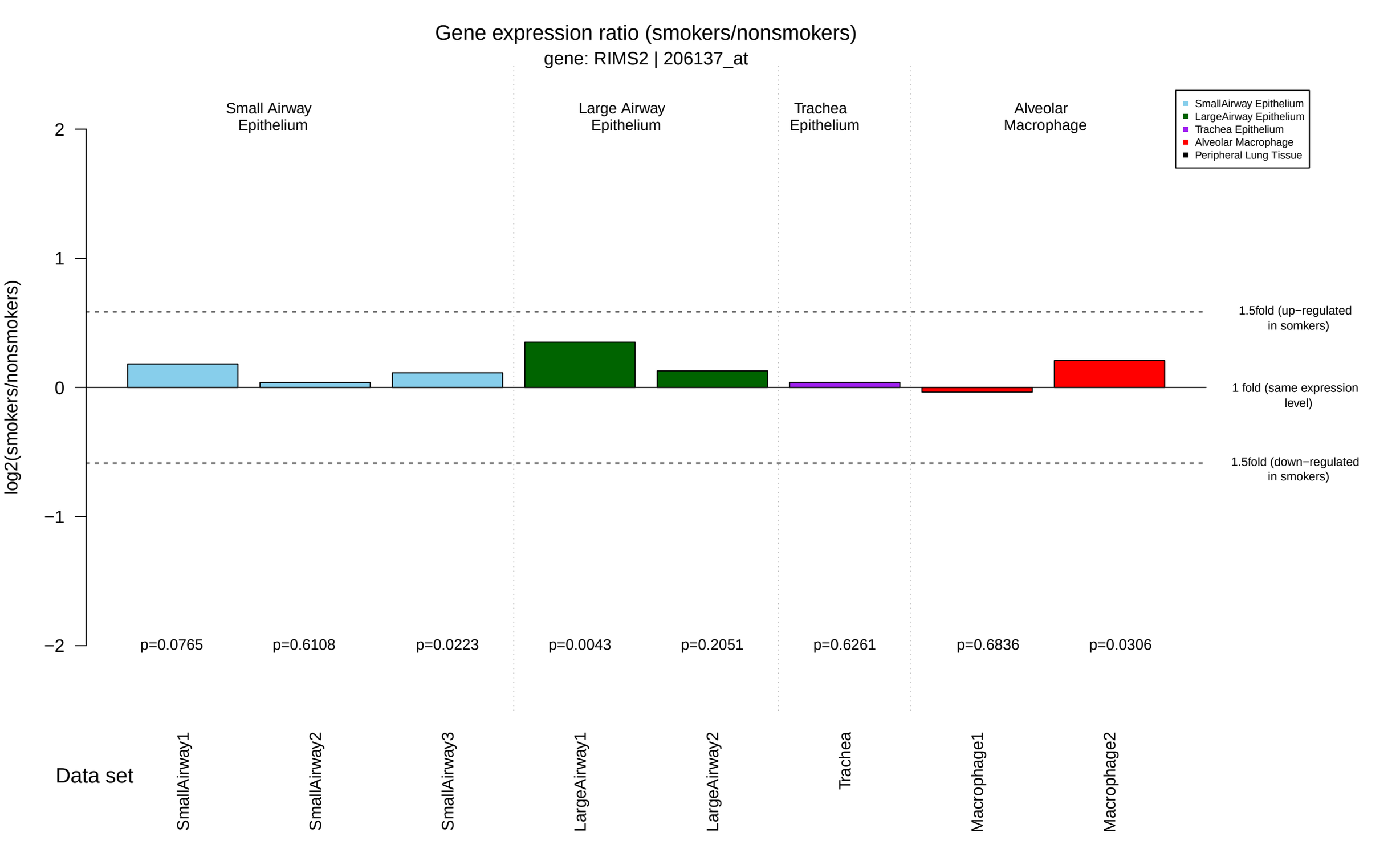
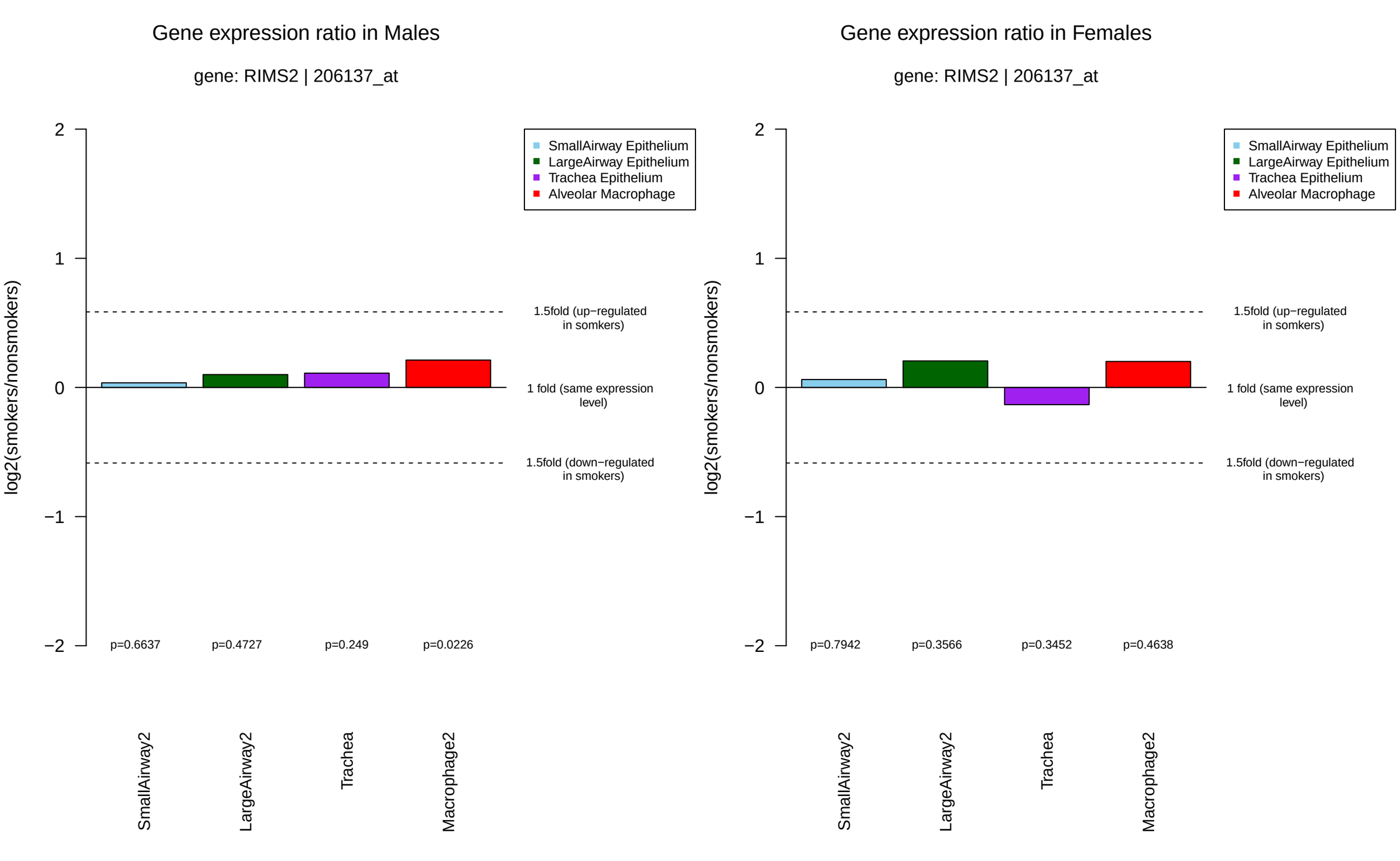
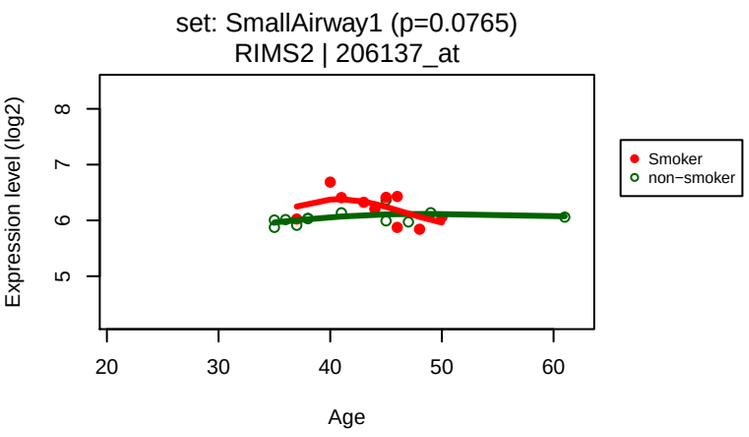
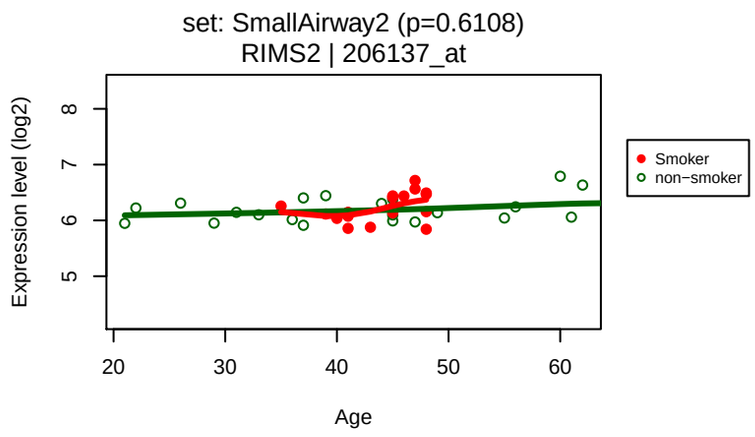
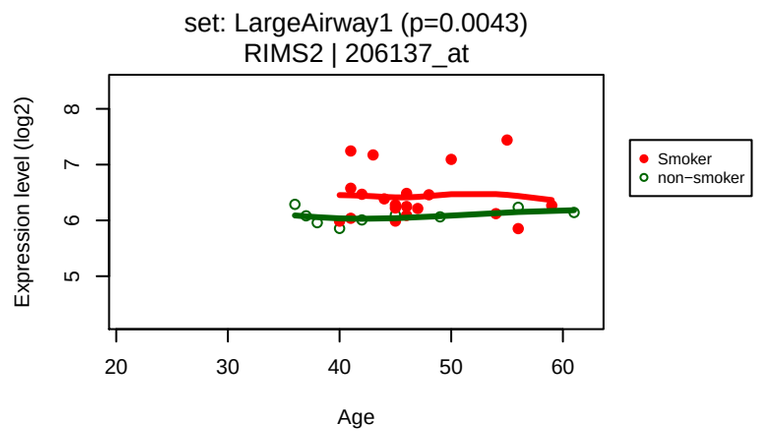

Gene expression of smokers and nonsmokers
gene: RIMS2 | 206137_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0765
p=0.6108
p=0.0223
p=0.0043
p=0.2051
p=0.6261
p=0.6836
p=0.0306
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

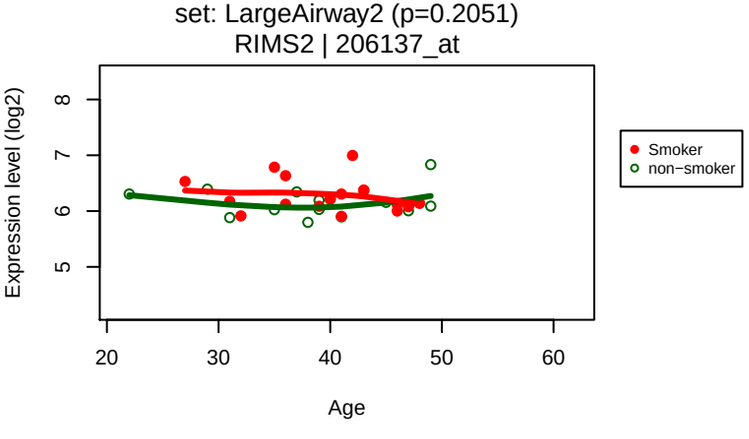

Gene expression ratio (smokers/nonsmokers)
gene: RIMS2 | 206137_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0765
p=0.6108
p=0.0223
p=0.0043
p=0.2051
p=0.6261
p=0.6836
p=0.0306
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

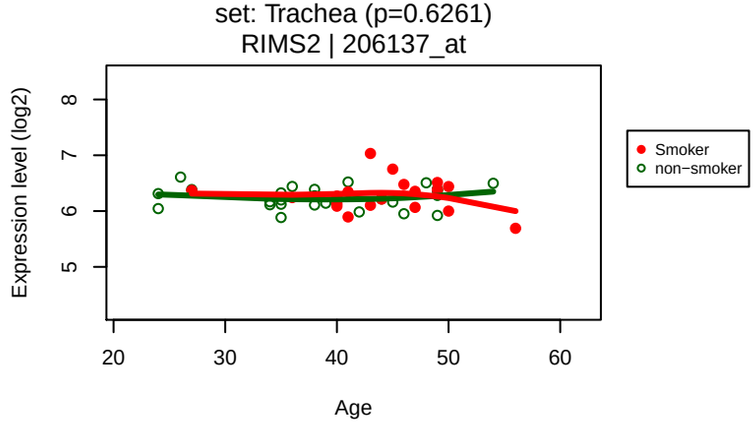

Gene expression ratio in Males
gene: RIMS2 | 206137_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6637
p=0.4727
p=0.249
p=0.0226
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RIMS2 | 206137_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7942
p=0.3566
p=0.3452
p=0.4638
SmallAirway2
LargeAirway2
Trachea
Macrophage2

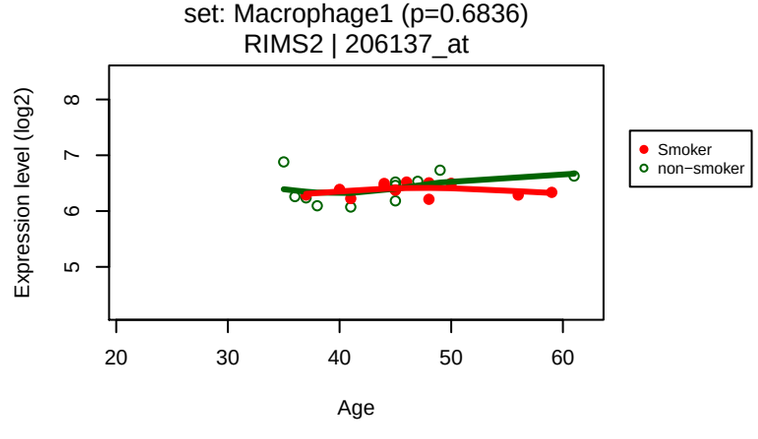

set: SmallAirway1 (p=0.0765)
RIMS2 | 206137_at
Expression level (log2)
Age
Smoker
non−smoker

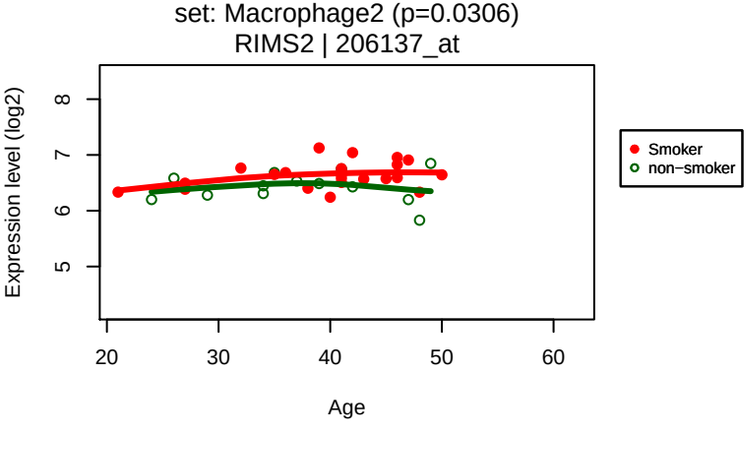

set: SmallAirway2 (p=0.6108)
RIMS2 | 206137_at
Expression level (log2)
Age
Smoker
non−smoker

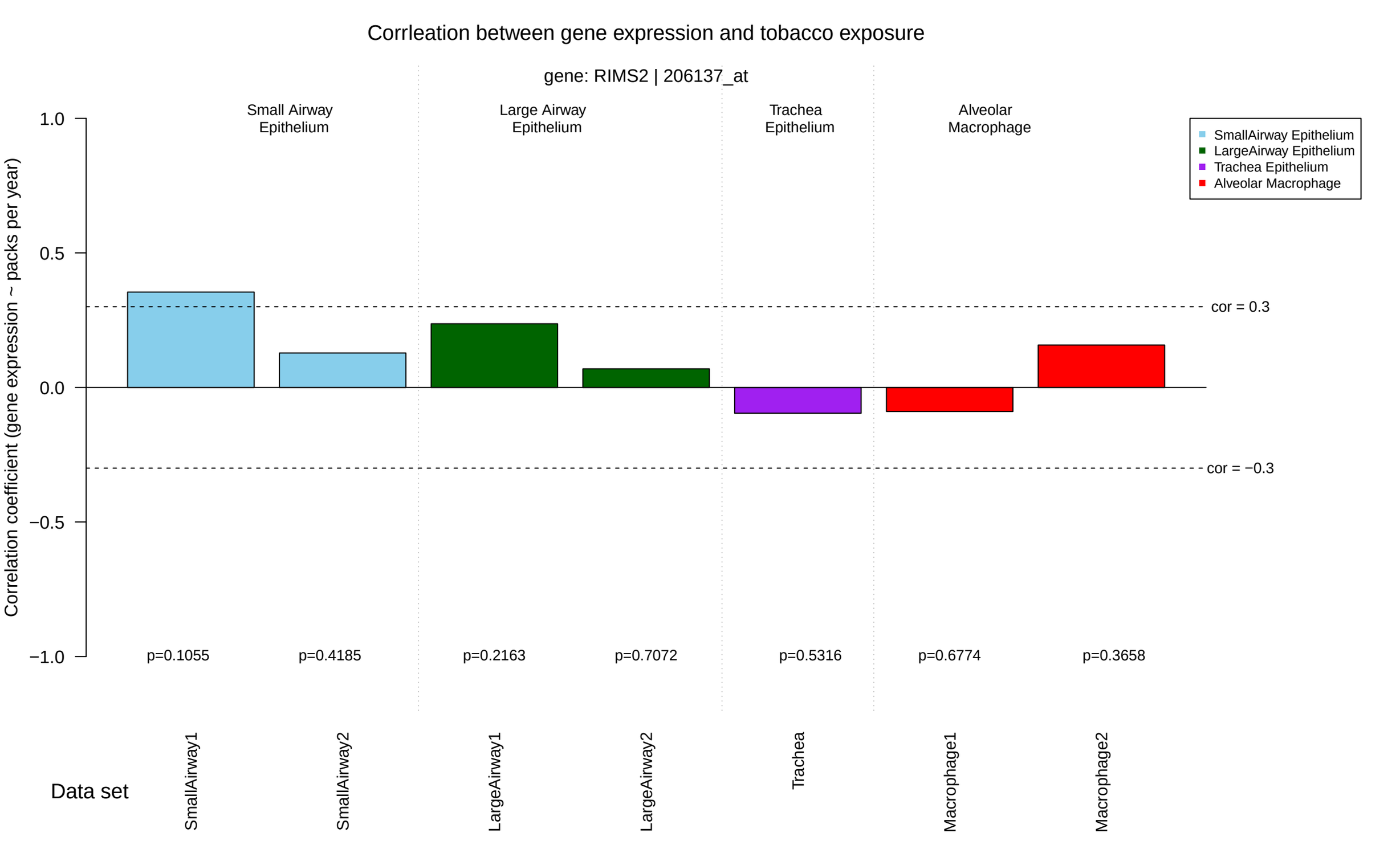

set: LargeAirway1 (p=0.0043)
RIMS2 | 206137_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2051)
RIMS2 | 206137_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6261)
RIMS2 | 206137_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6836)
RIMS2 | 206137_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0306)
RIMS2 | 206137_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RIMS2 | 206137_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1055
p=0.4185
p=0.2163
p=0.7072
p=0.5316
p=0.6774
p=0.3658
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2